

PRACTICAL WEEK-4

1. CREATE BOOKS TABLE

The screenshot displays a database IDE interface. The top pane shows SQL commands for creating a database and a table. The bottom pane shows the execution results, including a table structure grid and an action output log.

SQL Commands:

```
1 DROP DATABASE IF EXISTS bookflow_db;
2 CREATE DATABASE bookflow_db;
3 USE bookflow_db;
4
5 CREATE TABLE Books
6 (
7     book_id INT AUTO_INCREMENT PRIMARY KEY,
8     title VARCHAR(200) NOT NULL,
9     author VARCHAR(150) NOT NULL,
10    genre VARCHAR(100),
11    description TEXT,
12    published_year INT
13 );
14
15
16
```

Execution Results:

#	Time	Action	Message	Duration / Fetch
2	14:32:08	DROP DATABASE IF EXISTS bookflow_db	2 row(s) affected	0.125 sec
3	14:32:29	CREATE DATABASE bookflow_db	1 row(s) affected	0.031 sec
4	14:32:33	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:34:07	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected	0.062 sec

Table Structure (Result Grid):

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

Action Output Log:

#	Time	Action	Message
3	14:32:29	CREATE DATABASE bookflow_db	1 row(s) affected
4	14:32:33	USE bookflow_db	0 row(s) affected
5	14:34:07	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected
6	14:39:04	DESCRIBE Books	6 row(s) returned

2. INSERT BOOK DATA

```

17 • INSERT INTO Books
18 (
19     title,
20     author,
21     genre,
22     description,
23     published_year
24 )
25 VALUES
26
27 (
28     'The Martian',
29     'Andy Weir',
30     'Science Fiction',
31     'An astronaut struggles to survive alone on Mars.',
32     2011
33 ),
34
35 (
36     'Dune',
37     'Frank Herbert',

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	14:32:33	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:34:07	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected	0.062 sec
6	14:39:04	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
7	14:42:05	INSERT INTO Books (title, author, genre,...	10 row(s) affected Records: 10 Duplicates: 0 W...	0.016 sec

3. DISPLAY BOOKS

```

107 • SELECT
108     book_id,
109     title,
110     author,
111     genre,
112     published_year
113 FROM Books

```

manually get help for the current caret position or to toggle automatic help.

Result Grid

	book_id	title	author	genre	published_year
1		The Martian	Andy Weir	Science Fiction	2011
2		Dune	Frank Herbert	Science Fiction	1965
3		The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4		Pride and Prejudice	Jane Austen	Romance	1813
5		The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6		Foundation	Isaac Asimov	Science Fiction	1951
7		The Hobbit	J.R.R. Tolkien	Fantasy	1937
8		Enders Game	Orson Scott Card	Science Fiction	1985
9		Harry Potter and the Sorcerer's Stone	J.K. Rowling	Fantasy	1997
10		Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870

Books 2 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	14:34:07	CREATE TABLE Books (book_id INT AUTO_INCREMENT ...	0 row(s) affected	0.062 sec
6	14:39:04	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
7	14:42:05	INSERT INTO Books (title, author, genre, descriptio...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
8	14:44:06	SELECT book_id, title, author, genre, published_y...	10 row(s) returned	0.000 sec / 0.000 sec

4. ADD EMBEDDING COLUMN

116 • ALTER TABLE Books
 117 ADD COLUMN embedding JSON;
 118
 119 • DESCRIBE Books;
 120
 121
 122
 123

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Result 3 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	14:46:43	ALTER TABLE Books	0 row(s) affected	0.015 sec
10	14:46:49	ADD COLUMN embedding JSON	Error Code: 1064. You have an error in your SQL syntax; check...	0.000 sec
11	14:49:13	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
12	14:49:41	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec

5. INSERT MOCK VECTORS

1. BOOK 1 - MARTIN

121 • UPDATE Books
 122 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
 123 WHERE book_id = 1;
 124
 125
 126
 127

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	14:46:49	ADD COLUMN embedding JSON	Error Code: 1064. You have an error in your SQL syntax; check...	0.000 sec
11	14:49:13	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
12	14:49:41	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec
13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec / 0.000 sec

2. DUNE

```

125 • UPDATE Books
126 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
127 WHERE book_id = 2;
128
129
130

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 11	14:49:13	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 12	14:49:41	DESCRIBE Books	7 row(s) returned
✓ 13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

3. HITCHHIKERS GUIDE

```

129 • UPDATE Books
130 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
131 WHERE book_id = 3;
132
133

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 12	14:49:41	DESCRIBE Books	7 row(s) returned
✓ 13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

4. PRIDE AND PREJUDICE

```

133 • UPDATE Books
134 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
135 WHERE book_id = 4;
136
137
138

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 13	14:50:45	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	14:54:50	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

5. GREAT GATSBY

```

137 • UPDATE Books
138 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
139 WHERE book_id = 5;
140
141
142

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 14	14:52:06	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	14:54:50	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	14:55:40	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

6. FOUNDATION

```

141 • UPDATE Books
142 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
143 WHERE book_id = 6;
144
145
146

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 15	14:53:53	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	14:54:50	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	14:55:40	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	14:56:14	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

7. THE HOBBIT & ENDERS GAME

```

145 • UPDATE Books
146 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
147 WHERE book_id = 7;
148
149 • UPDATE Books
150 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
151 WHERE book_id = 8;
152
153

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 17	14:55:40	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	14:56:14	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 19	14:57:09	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	14:57:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

8. HARRY POTTER & TWENTY THOUSAND LEAGUES

```

153 • UPDATE Books
154   SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
155   WHERE book_id = 9;
156
157 • UPDATE Books
158   SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
159   WHERE book_id = 10;
160
161
162

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 19	14:57:09	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	14:57:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 21	15:01:30	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

9. VERIFY ALL VECTORS

```

161 • SELECT
162     book_id,
163     title,
164     embedding
165 FROM Books
166 ORDER BY book_id;
167

```

manually current c toggle

Result Grid

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
• NULL	NULL	NULL

Books 4 x Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
✓ 20	14:57:21	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 21	15:01:30	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 23	15:04:21	SELECT book_id, title, embedding FROM Books ORDE...	10 row(s) returned

10. L2 DISTANCE

168 • **SELECT**

169 book_id,

170 title,

171 author,

172 genre,

173 ROUND(
174 SQRT(

Result Grid | | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Andy Weir	Science Fiction	0
	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
	2	Dune	Frank Herbert	Science Fiction	0.0866

Result Grid
Form Editor
Field Types

Result 5 x | Read Only | Context Help

Output

Action Output

#	Time	Action	Message
✓ 21	15:01:30	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 23	15:04:21	SELECT book_id, title, embedding FROM Books ORDE...	10 row(s) returned
✓ 24	15:10:15	SELECT book_id, title, author, genre, ROUND(...	3 row(s) returned

11. COSINE SIMILARITY

215 • **SELECT**

216 book_id,

217 title,

218 author,

219 genre,

220 ROUND(
221 (
222 CAST(
223 JSON_EXTRACT(embedding, '\$[0]')

Result Grid | | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result Grid
Form Editor
Field Types

Result 6 x | Read Only | Context Help

Output

Action Output

#	Time	Action	Message
✓ 22	15:01:47	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 23	15:04:21	SELECT book_id, title, embedding FROM Books ORDE...	10 row(s) returned
✓ 24	15:10:15	SELECT book_id, title, author, genre, ROUND(...	3 row(s) returned
✓ 25	15:14:09	SELECT book_id, title, author, genre, ROUND(...	3 row(s) returned